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Roll No

MI-4002-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Drilling and Blasting

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe factors affecting performance of drilling in opencast mine.
b) Describe parameters for selection of drilling system.
2. Describe Cast Blasting with its merits.
3. Describe SMS system with its merits.
4. Explain in detail about
 - a) Ring drilling method for underground mines.
 - b) Rotary drilling method for underground mines.
5. Explain about Bulk explosives initiation systems and accessories.
6. a) Design a suitable blast hole pattern for a drift in sand stone having dimension 5×3 m.
b) What are role of cut holes, easers and trimmers in underground blast design?

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7. a) Describe the construction and working of an electronic detonator with a neat sketch.
b) What is meant by permitted explosives? Explain their properties and uses in U/G mines.

8. a) Design a blast for a Coal mine OB having a bench height of 10.0m. Assume your own conditions. Also estimate the powder factor and drill factor.
b) Define the following:
 - i) Coupling
 - ii) Decking
 - iii) Subgrade
